

Exercise 3 — Group Report

1. Goal

The goal of the required part was to build a writer-identification / writer-retrieval system. Each handwriting image is represented by local SIFT descriptors. These local descriptors are aggregated into one global image descriptor using VLAD. Finally, retrieval performance is measured using Top-1 accuracy and mean average precision (mAP).

2. Required Method:

1. Extract SIFT descriptors from each handwriting image.
2. Fix all SIFT keypoint orientations to 0.
3. Apply Hellinger normalization to the SIFT descriptors.
4. Learn a visual dictionary using MiniBatchKMeans.
5. Encode each image using VLAD.
6. Apply different normalization steps.
7. Evaluate retrieval using Top-1 accuracy and mAP.
8. Apply Exemplar-SVM as the final representation step.

Parameter Setting:

Parameter	Value
Codebook size K	100
Number of sampled descriptors	500000
PCA whitening components	512
Exemplar-SVM C	1000
Binary preprocessing	True

3. Results:

Method	Top-1 Accuracy	mAP
Plain VLAD	0.820556	0.629558
VLAD + power normalization	0.832500	0.644012
VLAD + power + intra-normalization	0.838889	0.649681
VLAD + power + intra + PCA whitening, 512 components	0.882778	0.744912
Exemplar-SVM on VLAD representation	0.895556	0.769657

4. Step-by-Step Improvement:

Comparison	mAP Before	mAP After	Change
Plain VLAD -> Power normalization	0.629558	0.644012	+0.014454
Power normalization -> Intra-normalization	0.644012	0.649681	+0.005669
Intra-normalization -> PCA whitening	0.649681	0.744912	+0.095231
PCA whitening -> Exemplar-SVM	0.744912	0.769657	+0.024745
Plain VLAD -> Final Exemplar-SVM	0.629558	0.769657	+0.140099

5. Interpretation

Plain VLAD already achieved a strong baseline result with mAP 0.629558. This shows that SIFT descriptors and VLAD are suitable for this writer-retrieval task.

Power normalization improved mAP from 0.629558 to 0.644012. This means that visual burstiness was present in the VLAD representation. Visual burstiness means that some local patterns occur very often and can dominate the similarity score. Power normalization reduces this effect.

Intra-normalization gave a smaller additional improvement, increasing mAP from 0.644012 to 0.649681. This shows that normalizing each visual-word block separately is useful, although the improvement was smaller than for power normalization.

PCA whitening gave the largest single improvement. It increased mAP from 0.649681 to 0.744912. This means that the VLAD dimensions contained correlated or redundant information. PCA whitening made the representation more compact and more discriminative.

Exemplar-SVM gave the best final result. It increased mAP from 0.744912 to 0.769657. This shows that learning a discriminative direction for each test image against the training images improved the retrieval representation.

6. Conclusion

The required SIFT + VLAD pipeline worked well. The performance improved almost step by step. The strongest improvement came from PCA whitening, while the best final result was obtained with Exemplar-SVM.

Final required result: Top 1 accuracy = 0.895556, mAP = 0.769657

This shows that the classical SIFT + VLAD pipeline, combined with normalization, PCA whitening, and Exemplar-SVM, is effective for writer retrieval on the ICDAR17 historical handwriting dataset.